

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0510 Slot A

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 10%

QUESTIONS:5 Total Marks 50 Duration :60 Minutes Annexure A
Comparative Analysis

Code of Conduct:

I M.Veeranjaneya Reddy(192472140) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.Veeranjaneya Reddy

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant	
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison	
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided	
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation	
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

1. Detection Performance Comparison

Two object detection models produce the following results:

Model True Positives False Positives False Negatives

A (Viola-Jones) 80 20 30

B (YOLO) 90 15 20

(a) Compute precision and recall for both models.

(b) Compare their performance and state which model is better for object detection.

Ans:

Given:

Model	TP	FP	FN
A – Viola-Jones	80	20	30
B – YOLO	90	15	20

(a) Compute precision and recall for both models.

Precision:

Recall:

Model A – Viola-Jones

Model B – YOLO

(b) Compare their performance and state which model is better.

Measure	Viola-Jones	YOLO
Precision	80%	85.71%
Recall	72.73%	81.82%

YOLO has higher precision, meaning it produces fewer false positive detections. It also has higher recall, meaning it detects more of the actual objects.

Therefore, **YOLO is better for object detection** because it performs better in both precision and recall.

2. Motion Estimation Error Analysis

Two motion estimation techniques produce displacement errors (in pixels):

Frame Method A Method B

1 2 3

2 3 5

3 2 4

(a) Compute average error for both methods.

(b) Compare and evaluate which method is more accurate.

Ans:

Given:

Frame	Method A	Method B
1	2	3
2	3	5
3	2	4

(a) Compute average error for both methods.

Average error is:

Method A:

Method B:

(b) Compare and evaluate which method is more accurate.

Method	Average Error
Method A	2.33 pixels
Method B	4.00 pixels

Method A has a lower average displacement error than Method B.

Thus, Method A has **1.67 pixels less average error**.

Since lower error indicates more accurate motion estimation, **Method A is more accurate**.

Conclusion: Method A provides better motion estimation accuracy based on the given displacement errors.

3. Optical Flow Magnitude Comparison

Optical flow magnitudes (in pixels/frame):

RegionMethod AMethod B

R1 5 8

R2 6 10

R3 5 9

(a) Compute average motion magnitude for both methods.

(b) Compare and evaluate which method captures stronger motion.

Ans:

Given:

Region	Method A	Method B
R1	5	8
R2	6	10
R3	5	9

(a) Compute average motion magnitude for both methods.

Average motion magnitude:

Method A:

Method B:

(b) Compare and evaluate which method captures stronger motion.

Method	Average Motion Magnitude
Method A	5.33 pixels/frame
Method B	9 pixels/frame

Method B has a higher magnitude in all three regions:

- R1: **8 > 5**
- R2: **10 > 6**
- R3: **9 > 5**

Difference in average magnitude:

Therefore, **Method B captures stronger motion** because its average optical-flow magnitude is higher.

Conclusion: Method B is better at capturing stronger motion in the given regions.

4. Depth Estimation Comparison (Stereo vs SfM)

Estimated depth values (in meters):

Object Stereo Vision Structure from Motion

A 2.0 2.5

B 3.0 3.8

C 4.0 5.0

Actual depths: A=2, B=3, C=4

(a) Compute absolute error for both methods.

(b) Compare and evaluate which method is more accurate.

Ans:

Given:

Object	Stereo Vision	SfM	Actual Depth
A	2.0 m	2.5 m	2 m
B	3.0 m	3.8 m	3 m
C	4.0 m	5.0 m	4 m

(a) Compute absolute error for both methods.

Formula:

Stereo Vision:

For A:

For B:

For C:

Average Stereo error:

Structure from Motion (SfM):

For A:

For B:

For C:

Average SfM error:

(b) Compare and evaluate which method is more accurate.

Method	Average Absolute Error
Stereo Vision	0 m
SfM	0.767 m

Stereo Vision has **zero error** for all three objects, while SfM has errors of 0.5 m, 0.8 m and 1.0 m.

Since a lower absolute error represents better depth estimation accuracy, **Stereo Vision is more accurate.**

Conclusion: Stereo Vision performs better for the given dataset because its average error is **0 m**, compared with **0.767 m** for SfM.

5. Pattern Recognition Accuracy Comparison

Two recognition systems produce results:

System Correct Predictions Total Samples

Traditional 85 100

Deep Learning 92 100

(a) Compute accuracy for both systems.

(b) Compare and evaluate which system performs better.

Ans:

Given:

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

(a) Compute accuracy for both systems.

Formula:

Traditional System:

Deep Learning System:

(b) Compare and evaluate which system performs better.

System	Accuracy
Traditional	85%
Deep Learning	92%

Difference:

Traditional system incorrect predictions:

Deep Learning incorrect predictions:

Therefore, Deep Learning makes **7 fewer incorrect predictions** out of 100 samples.

Conclusion: Deep Learning performs better because it achieves **92% accuracy**, compared with **85%** for the Traditional system.